TASK 3: 24V STARTING AND CHARGING SYSTEM DIAGNOSIS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 5 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 59-74 | Months 16-20

Your task

Investigate a heavy vehicle with slow cranking or poor battery charging. Test the 24 V battery set, starting circuit and alternator control, then identify the underlying fault and verify the result.

What you must do

1. Confirm symptoms, battery configuration, system voltage and technical data.
2. Inspect, isolate as required, and test both batteries under controlled conditions.
3. Measure cranking voltage/current and positive/ground voltage drop.
4. Test charging output, control and voltage at specified conditions.
5. Diagnose root cause, record repair action and complete post-repair verification.

What good evidence looks like

SAFE CONTROL

Control ignition/start authorisation, battery gases and high fault current; use insulated terminal protection and the specified disconnect sequence.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or 24 V starting/charging rig
- Wiring diagram and specifications
- Battery tester, multimeter and current clamp
- PPE, terminal protection and job card

SOW ALIGNMENT

Block 5 - 12/24V Electrical, Electronic and Networked Systems

KSB evidence focus: K3, S6, S9; reinforced K1, K4, K10, S1-S4, S8, S10-S11, B2-B3

Task reference: HV_L3_Block_5_Task_3_24V_Starting_and_Charging_System_Diagnosis

TASK 3: 24V STARTING AND CHARGING SYSTEM DIAGNOSIS

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Battery 1 / 2 voltage		Battery test result	
Cranking voltage / current		Cranking specification	
Positive / ground drop		Voltage-drop limits	
Charge voltage / current		Diagnosis / verification	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why must batteries in a 24 V series pair be assessed individually and together?

2. What does voltage drop reveal that resistance testing may not?

Knowledge check - continued

3. Explain safe control of high starter current and battery gases.

4. How would you distinguish an alternator fault from a control or wiring fault?

5. What verification proves the repair is effective?

Assessor/Workplace Mentor observation

- ☐ Identifies 24 V configuration and battery hazards.
- ☐ Follows wiring data and records test conditions.
- ☐ Completes repair verification and technical record.
- ☐ Uses battery, voltage-drop and current tests correctly.
- ☐ Uses results to identify root cause.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

☐ PASS ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 5 | Task 3

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. Parasitic drain

Checks: Confirm sleep mode and approved wait time; use a suitable low-current method and isolate suspect circuits without waking modules.

Evidence: Record key-off current, time/conditions, specification, source circuit and retest.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.